RAJARAMBAPU INSTITUTE OF TECHNOLOGY, RAJARAMNAGAR

(An Empowered Autonomous Institute, Affiliated to Shivaji University, Kolhapur)

Industry Internship & Project (IIP) Report

on

“Food Ordering System”

(Under “Industry Internship (II)” Track)

Submitted in partial fulfilment of the requirements for the award of the degree of

Master of Computer Applications (M.C.A.)

in

Master of Computer Applications

by

Krutika Tanaji Patil

2447068

at

ScaleFull Technologies ,Pune

Under the Guidance of

Mrs. Anita Khamitkar

Department of Computer Applications

K. E. Society’s

Rajarambapu Institute of Technology, Rajaramnagar

(An Empowered Autonomous Institute, Affiliated to Shivaji University, Kolhapur)

Academic Year: 2025–2026

Certificate

K. E. Society's

Rajarambapu Institute of Technology, Rajaramnagar

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Department of Computer Applications

CERTIFICATE

This is to certify that the Internship and Project work under Industry Internship (II) track completed at **"ScaleFull Technologies , Pune"** is the bona-fide work carried out by the following student of Rajarambapu Institute of Technology, Rajaramnagar during the academic year 2025–26, in partial fulfilment for the award of the degree of **Master of Computer Applications (M.C.A.)**.

The contents of this report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree.

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PRN: **2447068**

Date: __________________ Place: Rajaramnagar

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Declaration of Academic Honesty and Integrity

I hereby declare that this report reflects my understanding of the subject and is written in my own words. I have duly cited and referenced all the sources consulted during the preparation of this work. I confirm that this report has neither been plagiarized nor submitted for the award of any other degree or certification. I further declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented, fabricated, or falsified any idea, data, fact, or source in this submission.

I also declare that the content of this report is original and has not been generated using Artificial Intelligence tools. Any digital or AI-based tools, if used, were limited only to minor assistance such as grammar correction or language refinement, and not for generating the core content of the report. I understand that any violation of the above statements may lead to disciplinary action by the Institute.

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Date: ________________ Place: RIT, Rajaramnagar

Acknowledgement

I would like to express my sincere gratitude to all those who contributed to the successful completion of my Industry Internship & Project (IIP) under the Industry Internship (II) Track at **ScaleFull Technologies,Pune**. This internship gave me an opportunity to apply academic knowledge in a real software development environment and to understand the practical workflow followed in web application development.

I am deeply thankful to my college mentor, **Dr Manisha Pawar**, for her continuous guidance, support, and encouragement throughout the internship and project work. I also express my gratitude to **Mrs. Manisha Pawar**, Head of the Department, Department of Master of Computer Applications, Rajarambapu Institute of Technology, Rajaramnagar, for providing the necessary academic support and facilities. I am also thankful to **Mr. Ashish Kulkarni**, Training and Placement Coordinator, for the required departmental coordination and support.

I sincerely thank my industry mentor, **Mrs Anita Khamitkar**, for providing me guidance during the internship and for allowing me to work in the software development domain. I am also thankful to the team members of ScaleFull Technologies for their valuable suggestions and support during the project development.

Finally, I express my heartfelt gratitude to my family, friends, and faculty members for their constant support, motivation, and encouragement during the completion of this report.

Abstract

This report presents the work carried out during the Industrial Internship Program at ScaleFull Technologies LLP in the domain of Full Stack Web Development using MERN Stack technologies. The internship focused on practical implementation of frontend and web application development concepts using HTML5, CSS3, Bootstrap 5, JavaScript ES6, React.js, Node.js, Express.js, MongoDB, and GitHub.

During the internship period, various practical assignments, mini-projects, and implementation tasks were completed to improve technical understanding and software development skills. The work included responsive webpage development, JavaScript programming, DOM manipulation, CRUD operations, API integration, React.js application development, state management, and routing systems.

As part of the practical learning process, several web applications and modules were developed using the MERN stack architecture. These projects involved designing reusable React components, managing application state, implementing backend APIs with Express.js and Node.js, and storing application data in MongoDB databases. Team collaboration, task management, and problem-solving approaches followed in a professional development environment also enhanced communication and project-handling skills.

The internship significantly improved practical coding abilities, debugging skills, logical problem-solving capabilities, and understanding of professional software development workflows. The report presents detailed information about the company profile, weekly work summary, technologies learned, practical investigations, literature review, project implementation, results obtained, and conclusions derived during the internship period.

This report presents detailed information about the company profile, weekly work summary, technologies learned, practical investigations, literature review, project implementation, challenges faced during development, results obtained, and conclusions derived during the internship period. The internship experience proved highly beneficial in bridging the gap between academic learning and industry-oriented software development practices.

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1 Introduction

1.1 Introduction to Industry Internship

Industry Internship & Project (IIP) is an important part of the academic curriculum because it provides practical exposure to the software development process followed in industry.

The Industrial Internship Program (IIP) equips the students with the skills needed to become more proficient in technical skills, communication, teamwork, and problemsolving. This will help the students familiarize themselves with professional software development procedures. The internship was conducted in association with **ScaleFull Technologies**, Pune as a part of the Bachelor of Technology program in Computer Science and Engineering. The major scope of the internship involves the full stack web development process with MERN Stack technologies. During the internship tenure, hands-on experience was gained in frontend and backend technologies of web development through various tasks.

Some of the technologies covered during the internship were HTML5, CSS3, Bootstrap 5, JavaScript ES6, React.js, Node.js, Express.js, MongoDB, SQL, Git, and GitHub.

1.2-Internship Objectives

The major objectives of the Industrial Internship Program are as follows:

- Practical hands-on experience in software development technologies.
- Understanding of contemporary full stack web development principles and practices.
- Enhanced knowledge in MERN stack technologies.
- Creation of responsive and easy-to-use web applications.
- Enhanced coding, debugging, and problem-solving abilities.
- Understanding of version control with the use of Git and GitHub.
- Practical experience in developing React.js applications.
- Understanding of APIs and performing CRUD operations.
- Communication and teamwork skills enhancement.

1.3-Motivation of Present Work

The motivation behind this work was to understand how a real business application is planned and implemented in a professional software development environment. Many organizations need a structured system to manage customers, leads, users, forms, tasks, and submissions. Manual methods such as spreadsheets or paper-based records are difficult to maintain when the amount of data increases.

The foodOrdering system selected for this internship provides an opportunity to understand practical frontend development, role-based screens, dynamic forms, and database-driven application flow. It also helped me learn how the MERN stack can be used to build a scalable web application. The project was useful from an academic point of view as well as from an industry point of view

because it connected classroom learning with practical implementation

1.4-Need of Industry Internship

Industry internship is necessary because it helps students develop professional discipline, technical confidence, and problem-solving ability. It also helps students understand how tasks are assigned, how requirements are discussed, and how project modules are reviewed. During this internship, I learned the importance of writing clean code, preparing reusable components, testing user flow, and maintaining proper documentation.

1.5-Overview of Report

This report is prepared according to the IIP report structure provided by the department. It includes introduction, company background and structure, weekly jobs summary, technical contents, findings and recommendations, conclusion, references, plagiarism report page. The internship and project work are presented together in a continuous manner because the internship work and project development were combined as part of the same IIP activity

2.Company Background and Structure

2.1.Company Background

ScaleFull Technologies LLP is an Indian software development and IT solution company located in Pune, Maharashtra, India. The company has specialized in the following fields: Web Development, Web Application Development, E-commerce Solutions, Mobile Application Development, IT Consultancy, Recruitment, and Technical Training.

The firm mainly concentrates on providing high-level digital solutions and implementing learning methods for practical purposes. The firm offers industry-related technical training and internships to students and freshers.

The intern experience at the organization is designed to be learning-based, project-driven, team-based, and current with modern software development trends**.**

2.2.Company Work Structure

ScaleFull Technologies LLP follows a project-based and team-oriented work structure. The company is divided into departments such as Development, Training & Internship, Project Management, HR & Recruitment, and Client Support.

The development team works on web, mobile, and software projects using modern technologies. Interns and trainees receive practical learning through real-time projects, assignments, and mentor guidance.

The company follows a systematic workflow including requirement analysis, planning, development, testing, deployment, and maintenance. Team collaboration, technical learning, and client satisfaction are important parts of the organization's work culture.

2.3. Internship Details

Particular	**Details**
Student Name	Krutika Tanaji Patil
PRN / Roll No.	2447068
Department	Master of Computer Applications
Company Name	ScaleFull Technologies LLP
Company Location	Pune
Internship Mode	Offline
Internship Duration	15 January 2026 to 15 July 2026
Role	React Frontend Developer Intern
College Mentor	Mrs. Manisha Pawar
Industry Mentor	Mrs. Anita Khamitkar
Project Title	Food Ordering System
Technology Stack	MERN Stack

2.4. Role and Responsibilities

During the internship, my role was React Frontend Developer Intern. My responsibilities were mainly related to designing and developing frontend screens, creating reusable components, improving the user interface, testing form behavior, understanding API-based data flow, and preparing documentation.

The important responsibilities completed during the internship were:

1. Understanding project requirements and user roles.
2. Designing React-based login, dashboard, form, and lead management screens.
3. Creating reusable UI components such as buttons, forms, sidebars, tables, and modals.
4. Supporting frontend-backend API integration and checking response data.
5. Verifying MongoDB collections using MongoDB Compass.
6. Preparing report content, project screenshots, and final presentation material.

2.5. Project Overview

The project developed during the internship is a MERN Stack Food Ordering System. The application is designed to provide an online platform where users can browse food items, add products to the cart, place orders, and make online payments easily. The system helps in managing food ordering services digitally with an interactive and user-friendly interface.

The developed system includes separate functionalities for Admin and User roles. The Admin can manage food items, orders, users, and delivery status, while Users can register, log in, explore menu items, manage carts, and place orders online.

The system contains modules for user authentication, food item management, cart management, order placement, payment integration, order tracking, and admin dashboard management. The frontend of the project is developed using React.js, while Node.js and Express.js are used for backend development. MongoDB is used as the database to store information related to users, food items, orders, cart details, and payment records.

2.6.Internship Guidance and Review Process

During the internship, regular guidance helped me understand how real-world web applications are developed using MERN Stack technologies. The guidance process helped me understand that software development should follow a proper workflow including planning, frontend development, backend integration, database management, testing, and deployment.

The industry mentor guided me in understanding practical concepts related to React.js frontend development, API integration, MongoDB database handling, authentication systems, and project structure management. The college mentor supported the academic documentation, report preparation, and presentation activities throughout the internship period.

The review process mainly focused on the following points:

1. Understanding the project requirements before starting implementation.
2. Designing responsive frontend pages with proper layout and navigation.
3. Developing backend APIs and integrating them with the frontend.
4. Testing login, cart, payment, and order management functionalities.
5. Improving project output after feedback and preparing documentation with screenshots.

2.7. Internship Progress Details

The internship at ScaleFull Technologies LLP, Pune is currently ongoing as part of the Industrial Internship Program. During the internship period, practical training and project development activities are being carried out in the domain of MERN Stack Web Development. The internship work includes frontend and backend development, database management, API integration, testing, and project implementation using modern web technologies.

The internship provides practical exposure to real-time software development processes, teamwork, problem-solving techniques, and professional project handling. The ongoing work mainly focuses on developing a Food Ordering System using React.js, Node.js, Express.js, and MongoDB.

The internship experience is helping in improving technical skills, understanding industry-level development practices, and gaining knowledge about project workflow, debugging, and deployment processes. The details included in this report are based on the tasks, project modules, technical learning, and practical work completed during the internship period

3.Weekly Jobs Summary

3.1-Weekly Work Summary

The following table summarizes the weekly tasks completed during the internship. The weekly summary includes learning activities, technical practice, project development work, and documentation.

Table 3.1: Weekly Jobs Summary

Week	Technologies / Topics Learned	Work Done / Activities Performed
1	Introduction to MERN Stack and HTML5	Learned about MERN Stack architecture, basic HTML structure, tags, forms, tables, semantic elements, and webpage design. Created simple static webpages.
2	CSS3 Basics and Responsive Design	Learned CSS selectors, Box Model, spacing, colors, Flexbox, positioning, and responsive webpage styling. Designed responsive webpage sections.
3	CSS Grid, Git, and GitHub	Studied CSS Grid layouts, Git commands, and GitHub repository management. Created frontend mini projects and uploaded them to GitHub.
4	Advanced CSS and UI Design	Learned transitions, animations, hover effects, and responsive UI concepts. Designed interactive portfolio and landing page layouts.
5	Bootstrap 5 Basics	Learned Bootstrap Grid System, containers, cards, forms, buttons, and navigation bars. Developed responsive webpages using Bootstrap.
6	JavaScript Basics	Learned JavaScript syntax, variables, operators, conditions, loops, and functions. Created mini projects using JavaScript concepts.
7	JavaScript DOM Manipulation	Studied DOM manipulation, events, form validation, and dynamic webpage interaction. Built Password Generator and Validation projects.
8	Advanced JavaScript and API Concepts	Learned array methods, objects, asynchronous JavaScript, fetch API, and API integration. Built Weather Application using APIs.
9	React.js Basics	Learned JSX, components, props, state, and React project structure. Developed simple React applications and reusable components.
10	React Router and State Management	Learned React Router DOM, navigation, hooks, and component communication. Built multi-page React applications.
11	React API Integration	Studied useEffect Hook, API integration, dynamic routing, and data rendering. Started development of MERN Stack Food Ordering System frontend.
12	Node.js and Express.js	Learned backend development concepts, Express routing, middleware, REST APIs, and server creation. Built backend APIs for project modules.
13	MongoDB and Database Integration	Learned MongoDB database operations, collections, CRUD operations, and Mongoose schema design. Connected backend APIs with database
14	Authentication and Authorization	Implemented user login, registration, JWT authentication, password encryption, and protected routes in MERN project.

Week	Technologies / Topics Learned	Work Done / Activities Performed
15	Food Ordering System Development	Developed modules for food items, cart management, order placement, admin dashboard, and order tracking features.
16	Project Testing and Deployment	Performed project testing, debugging, API checking, frontend-backend integration, and GitHub project management. Worked on deployment and final project improvements.

3.2-Skills Developed During Internship

During the internship, I improved my knowledge of MERN Stack development, frontend and backend integration, responsive UI design, API handling, MongoDB database management, and project deployment. Along with technical learning, I also developed a better understanding of how real-world web applications are designed, developed, tested, and maintained using modern software development practices.

The major skills developed during the internship are as follows:

1. Improved ability to design responsive and interactive user interfaces using React.js, HTML5, CSS3, Bootstrap, and JavaScript.
2. Developed reusable React components for login pages, navigation bars, food item cards, cart sections, order pages, and admin dashboard modules.
3. Practiced React Hooks, state management, routing, conditional rendering, and dynamic data handling in frontend applications.
4. Learned how frontend applications communicate with backend APIs using Axios and REST API concepts.
5. Understood backend development using Node.js and Express.js for creating APIs, handling routes, and managing server-side operations.
6. Gained hands-on experience in MongoDB database operations including collections, CRUD operations, schema creation, and database connectivity using Mongoose.
7. Implemented user authentication and authorization features using JWT authentication and protected routes.
8. Improved debugging skills by solving frontend errors, API integration issues, database connection problems, and routing errors.
9. Learned the importance of Git and GitHub for version control, project management, code updates, and collaboration.
10. Improved professional communication, teamwork, problem-solving ability, and confidence in explaining project workflow, architecture, and implementation during reviews and presentations.

3.3- Internship Learning Summary

The weekly work helped me gradually progress from basic web development concepts to full MERN Stack project implementation. The initial weeks focused on HTML5, CSS3, JavaScript, Bootstrap, Git, GitHub, React.js basics, and responsive UI design. Later weeks focused on backend development using Node.js and Express.js, MongoDB database integration, API development, authentication, and frontend-backend connectivity.

The internship mainly involved the development of a MERN Stack Food Ordering System. Different modules such as user authentication, food item management, cart functionality, order placement, payment integration, admin dashboard, and order tracking were implemented during the project development process.

This step-by-step learning approach helped me understand both the technical and professional aspects of full stack web application development, including project workflow, debugging, testing, database handling, API integration, version control, and teamwork.

3.4-Detailed Internship Activity Record

Apart from weekly learning, the internship also involved repeated practical activities that helped improve my confidence in development work. The tasks were not limited to writing code; they also included understanding requirements, checking UI output, testing screen behavior, verifying database entries, and preparing documentation. These activities helped me understand the complete cycle followed in a software project.

Table 3.4: Detailed Internship Activity Record

Activity Area	Details of Work Performed
Requirement Study	Understood the purpose and workflow of the MERN Stack Food Ordering System, including user authentication, food management, cart functionality, order placement, and admin operations before implementation.
Frontend Design	Designed responsive login pages, home pages, food item sections, cart pages, order pages, and admin dashboard interfaces using React.js, HTML5, CSS3, and Bootstrap.
Component Development	Created reusable React components such as Navbar, Footer, Food Cards, Cart Items, Buttons, Forms, and Dashboard sections for better code management and reusability.
API Integration	Connected frontend screens with backend APIs using Axios and REST API concepts for user login, food data fetching, cart management.

Activity Area	Details of Work Performed
Backend Development	Developed backend functionalities using Node.js and Express.js including routing, middleware, authentication, and API handling.
Database Management	Worked with MongoDB and Mongoose for storing and managing users, food items, orders, cart details, and payment-related records.
Testing& Debugging	Checked frontend responsiveness, API responses, routing, authentication flow, cart operations, and database connectivity. Fixed UI alignment issues and runtime errors during testing.
Version Control	Used Git and GitHub for project management, code updates, repository handling, and maintaining version control during development.
Documentation	Collected project screenshots, prepared weekly summaries, documented technical concepts, and organized internship report content for submission.

The above activities helped me understand that MERN Stack project development is a continuous improvement process involving frontend design, backend logic, database handling, API integration, testing, debugging, and documentation. The internship experience improved both my technical knowledge and practical understanding of real-world software development practices.

4.Technical Contents

4.1 Problem Identification

Many food businesses and restaurants face difficulties in managing customer orders, menu items, payments, and delivery processes manually. Traditional food ordering methods are time-consuming, difficult to track, and may lead to order management errors, delayed service, and poor customer experience. Customers also expect a fast and convenient online platform for browsing food items and placing orders from anywhere.

The identified problem was to develop a web-based MERN Stack Food Ordering System that can manage food items, customer orders, cart operations, user authentication, and payment processes efficiently. The system should provide a user-friendly interface for customers to explore menu items and place orders easily, while also allowing the admin to manage food products, orders, and delivery status in a structured and organized manner.

4.2 Problem Statement

To design and develop a web-based Food Ordering System using the MERN Stack that allows users to securely register, log in, browse food items, manage carts, place orders, and make online payments through an interactive and responsive interface. The system should also provide an admin panel to manage food items, customer orders, users, and delivery status efficiently.

4.3 Project Objectives

The main objectives of the MERN Stack Food Ordering System are:

1. To provide secure user registration and login functionality for customers and admin users.
2. To allow users to browse available food items with a clean and responsive interface.
3. To enable users to add food items to the cart and manage their orders easily.
4. To provide a smooth order placement system with proper order confirmation.
5. To integrate online payment functionality for secure transactions.
6. To allow the admin to manage food items, including add, update, and delete operations.
7. To enable the admin to manage customer orders and update delivery status.
8. To store and manage all application data efficiently using MongoDB.
9. To develop a responsive and user-friendly frontend using React.js for better user experience.

4.5:User Roles in the System

Table 4.1: User Roles and Responsibilities

Role	Responsibilities
Admin	Manage food items, update menu details, monitor customer orders, update delivery status, and control overall system operations
User	Register and log in to the system, browse food items, add products to cart, place orders, make payments, and track order details.

4.6:Literature Review of Project

Online Food Ordering Systems are widely used by restaurants and food businesses to manage customer orders, menu items, payments, and delivery processes efficiently. These systems help customers browse food items, place orders online, and make secure payments through web applications. Modern food ordering applications are generally web-based and provide better accessibility, faster service, and improved customer experience.

Web-based food ordering systems are useful because they can be accessed from different devices using a web browser. They reduce manual work, improve order management, and provide centralized control over customer and food-related data. In this project, a web-based approach was selected because it offers easy accessibility, better scalability, and smooth integration between frontend and backend technologies.

React.js is a component-based JavaScript library used for developing interactive and dynamic user interfaces. It allows developers to build reusable components for webpages and manage application state efficiently. In the Food Ordering System, React.js was used to create login pages, home pages, food item sections, cart pages, order pages, and admin dashboard interfaces. The component-based architecture improved code reusability and frontend maintainability.

Node.js provides a JavaScript runtime environment for server-side development, while Express.js is used to create APIs, manage routes, and handle backend operations. In this project, Node.js and Express.js were used to develop backend APIs for user authentication, food item management, cart handling, order processing, and payment integration.

MongoDB is a NoSQL document-oriented database that stores data in flexible JSON-like documents. It is suitable for web applications that require efficient handling of dynamic and scalable data. The Food Ordering System uses MongoDB collections such as users, food items, carts, orders, and payments for storing application data.

JWT (JSON Web Token) authentication is commonly used in modern web applications to provide secure login and protected routes. After successful authentication, a token is generated and used for accessing authorized application features.

4.8 Role of Online Food Ordering System

An online food ordering system helps customers order food conveniently through a web application without visiting the restaurant physically. Traditional ordering methods may create issues such as delayed service, manual order handling, and difficulty in managing customer requests. A digital food ordering platform reduces these problems by providing an organized and automated ordering process.

In this project, the Food Ordering System allows users to browse food items, view prices, add products to the cart, and place orders easily. The system provides a responsive and user-friendly interface that improves customer experience and simplifies food ordering operations.

4.9:Requirement Analysis

Requirement analysis was the first important activity before developing the MERN Stack Food Ordering System. The project required a web-based application where users could browse food items, manage carts, place orders, and make payments through a responsive and user-friendly interface. The main requirement was to design a clean frontend and a secure backend system that could efficiently manage customer and admin operations.

The system required features such as user registration and login, food item display, cart management, order placement, payment integration, and order tracking. An admin panel was also required to manage food items, customer orders, and delivery status.

Table 4.2: Functional Requirements of Food Ordering System

Requirement	Description
User Authentication	The system should allow users and admin to securely register and log in to the application.
Food Item Management	Admin should be able to add, update, delete, and manage food items and menu details
Food Browsing	Users should be able to browse available food items with images, prices, and descriptions.
Cart Management	Users should be able to add food items to the cart, update quantities, and remove products from the cart.
Order Management	The system should allow users to place orders and view their order details.
Database Storage	Data should be stored in proper MongoDB collections.

Table 4.3: Non-Functional Requirements of Food ordering System

Requirement	Description
Usability	The interface should be simple and understandable for users.
Responsiveness	The system should work properly on desktop, laptop, tablet, and mobile screen resolutions..
Maintainability	Components and layouts should be reusable and easy to modify.
Security	User authentication and protected routes should be implemented to secure user and admin access..
Performance	The application should load pages smoothly and handle API responses efficiently without unnecessary delay.
Reliability	The system should manage food orders, cart operations, and database transactions accurately without data loss.

4.10: Technology Stack

The Food Ordering System was developed using the MERN Stack. The frontend was developed using React.js for creating interactive and responsive user interfaces. The backend was developed using Node.js and Express.js for handling APIs, routing, and server-side operations. MongoDB was used as the database for storing user, food item, cart, and order information. JWT authentication was implemented for secure login and protected access.

Development tools used during the project included Visual Studio Code, Git, GitHub, MongoDB Compass, and Postman for coding, version control, database management, and API testing.

Table 4.4: Technology Stack Used

Technology / Tool	Purpose
React.js	Frontend user interface development
JavaScript	Application logic and frontend functionality
HTML5 and CSS3	Webpage structure and styling
Bootstrap 5	Responsive UI design and styling
Node.js	Server-side runtime environment
Express.js	Backend API development and routing
MongoDB	Database for storing users, food items, carts, and orders
Mongoose	Database schema modeling and MongoDB connectivity
JWT	Token-based authentication and secure access
Axios	API communication between frontend and backend
MongoDB Compass	Database visualization and verification
Postman	API testing and debugging

Technology / Tool	Purpose
Git and GitHub	Version control and project management
VS Code	Code editor used for project development

4.11: System Architecture

The CRM system follows a client-server architecture. The React frontend communicates with the backend APIs. The backend processes requests, verifies authentication, applies business logic, and stores or retrieves data from MongoDB.

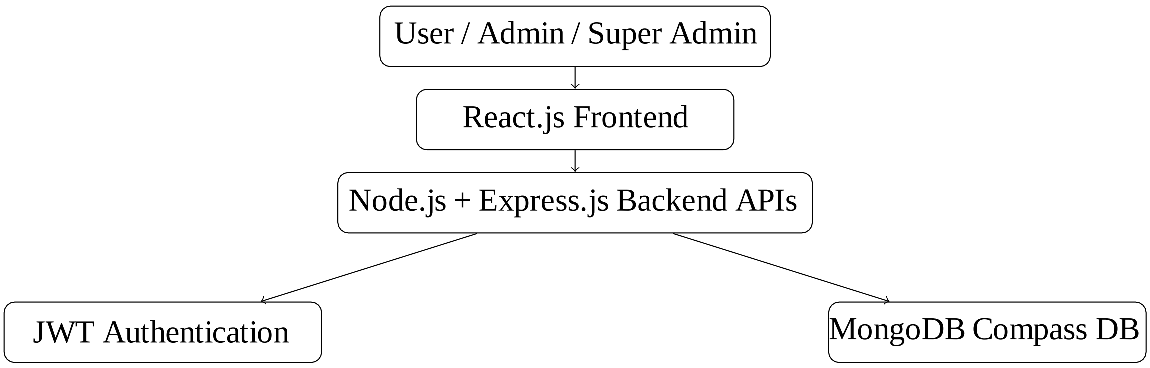


Figure 4.11: System Architecture of Food OrderingApplication

4.12: Database Design

The database used for the project is MongoDB. The database stores all application-related data including user information, food items, cart details, orders, and payment records. MongoDB was selected because it provides flexible document-based storage and efficient handling of dynamic web application data.

Table 4.5: MongoDB Collections Used in Food Ordering System

Collection	Purpose
users	Stores customer account details such as name, email, password, and login information.
fooditems	Stores food product details including name, category, price, image, and description.
carts	Stores cart items selected by users before order placement.
orders	Stores customer order details, order status, payment status, and delivery information.
payments	Stores online payment transaction details and payment records.
admins	Stores administrator login credentials and admin-related information.

4.13: Module 1: Login and Authentication

The login module provides secure access to the system. The system supports Admin, and User access. JWT authentication is used to manage login sessions and protected routes.

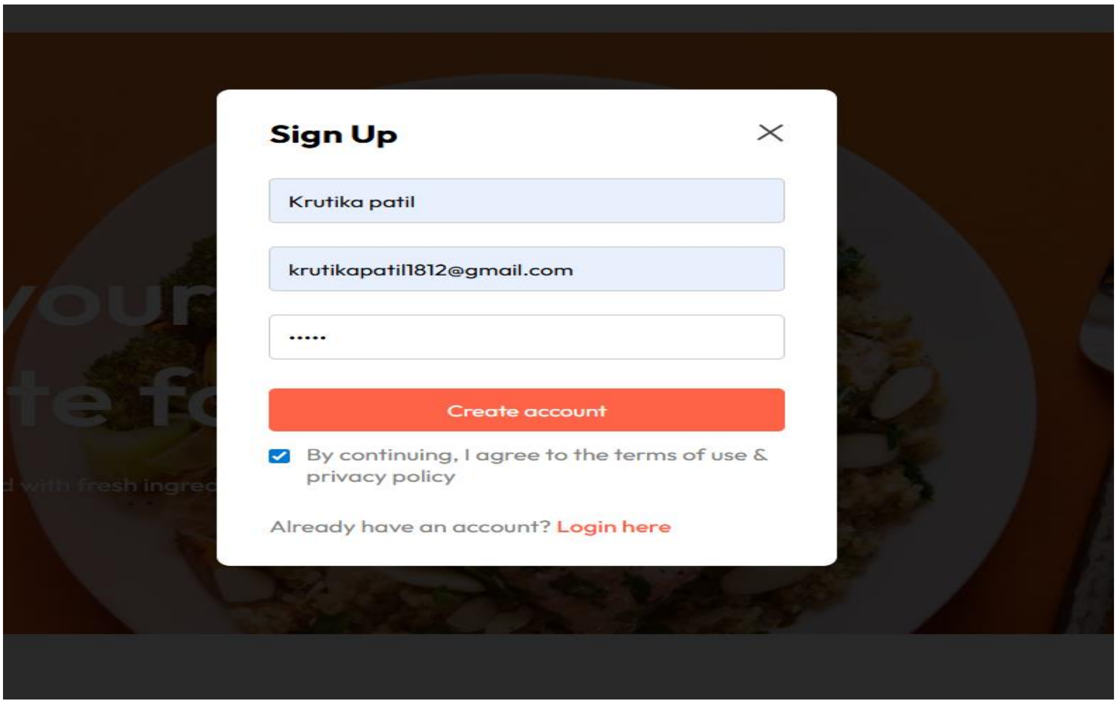


Figure1.1 : Users SignUp Screen

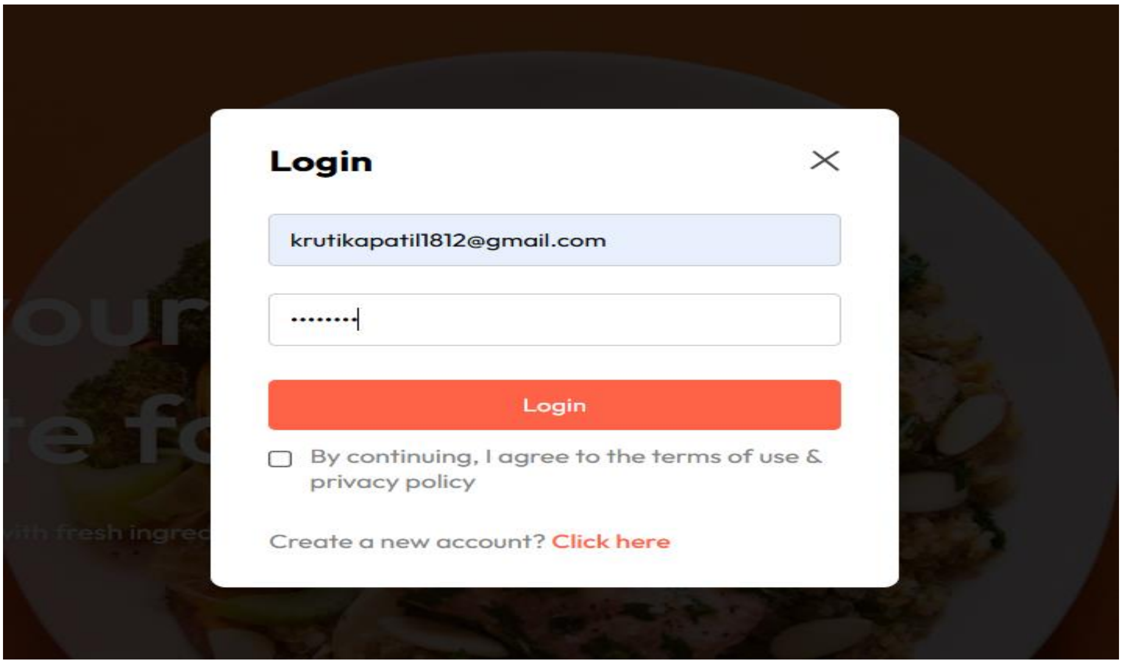
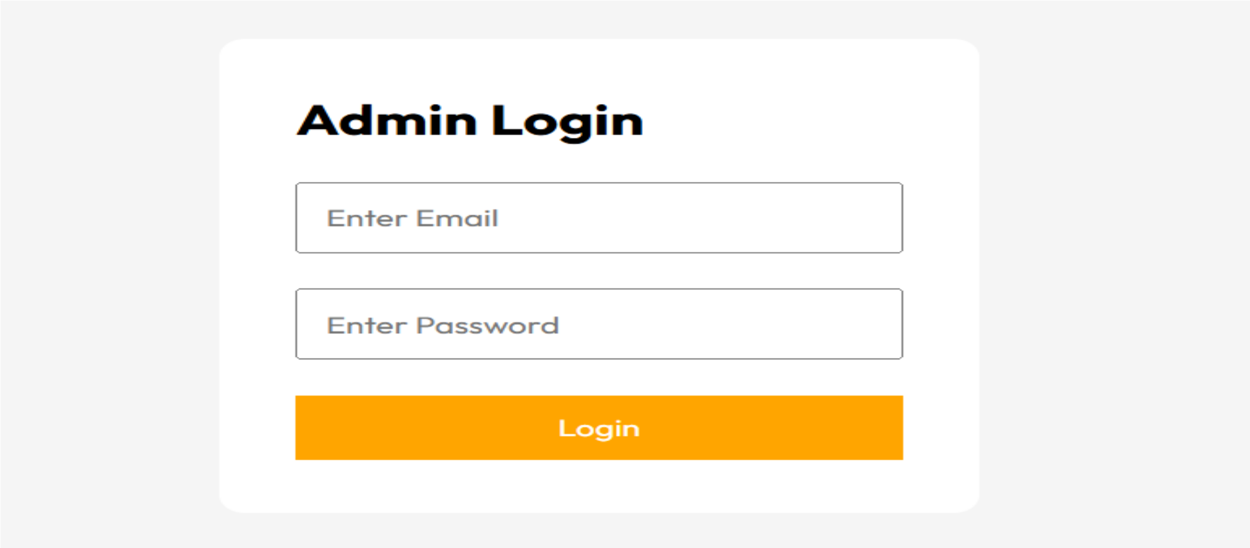
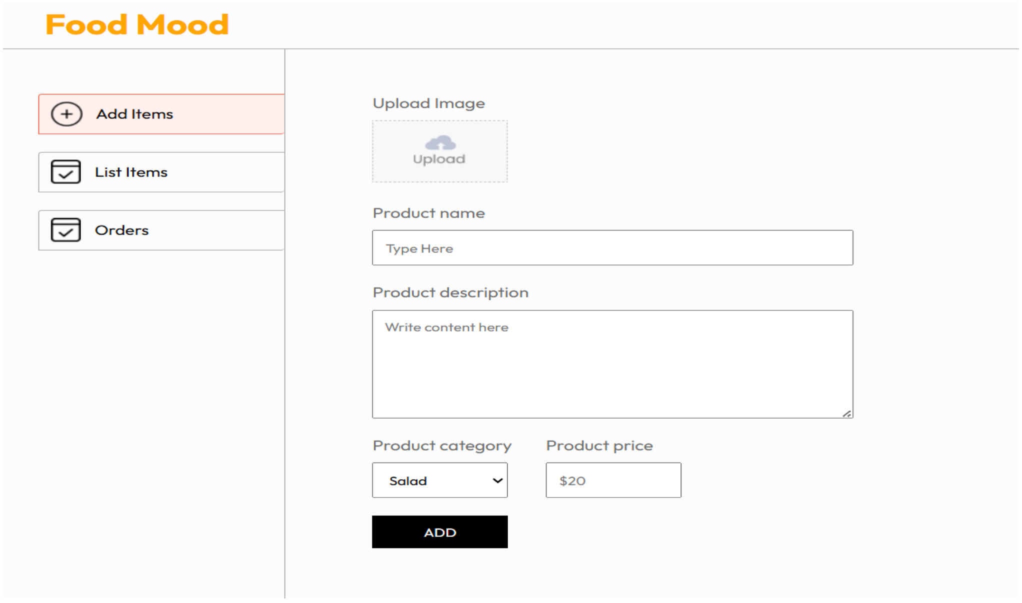


Figure 1.2 : Users Login Screen

4.14: Module 2: Admin Management

Admin management allows creation and monitoring of admin users. The screen contains a form for creating administrators and a directory table for managing existing administrators.

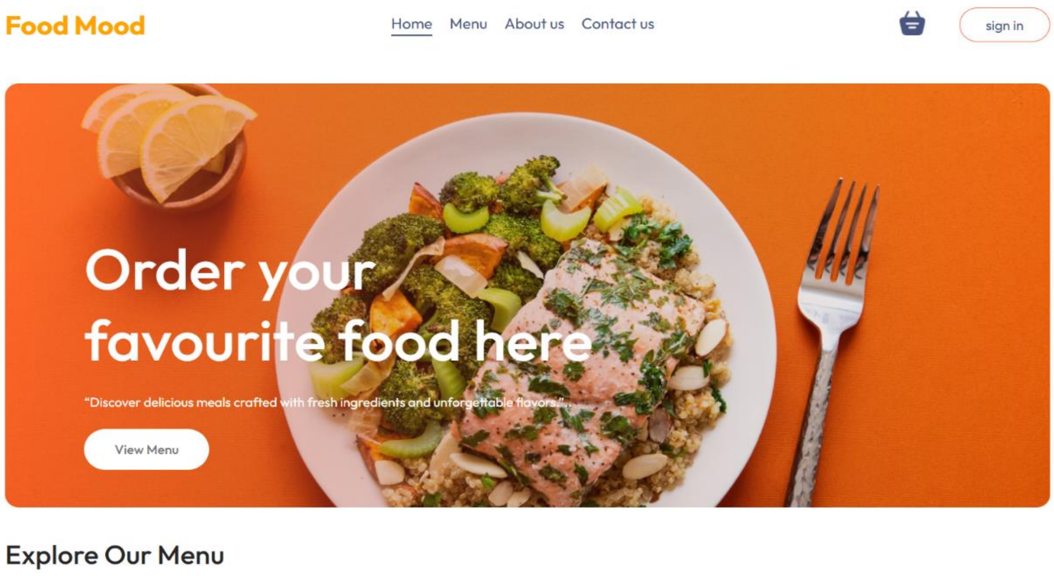




2.1Figure: Admin Management Screen

4.15: Module 3: Dashboard Management

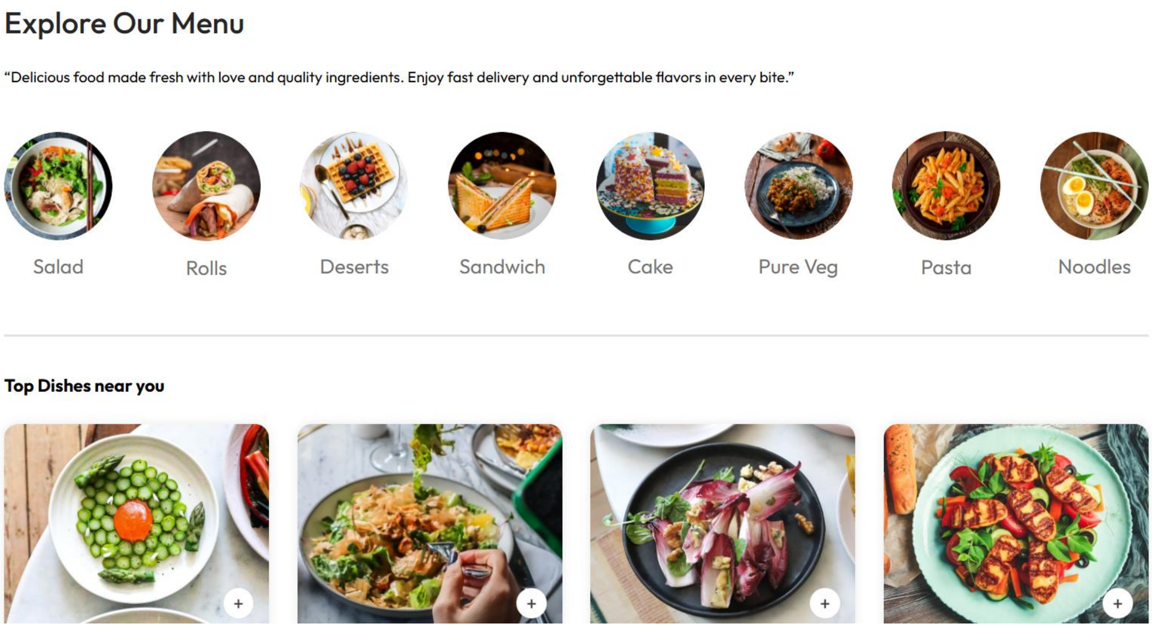
Dashboard management provides an overview of the complete Food Ordering System. The dashboard contains different sections for monitoring food items, customer orders, users, revenue details, and delivery status.



3.1 **Figure**:DashBoard Screen

4.16: Module 4: Cart Management

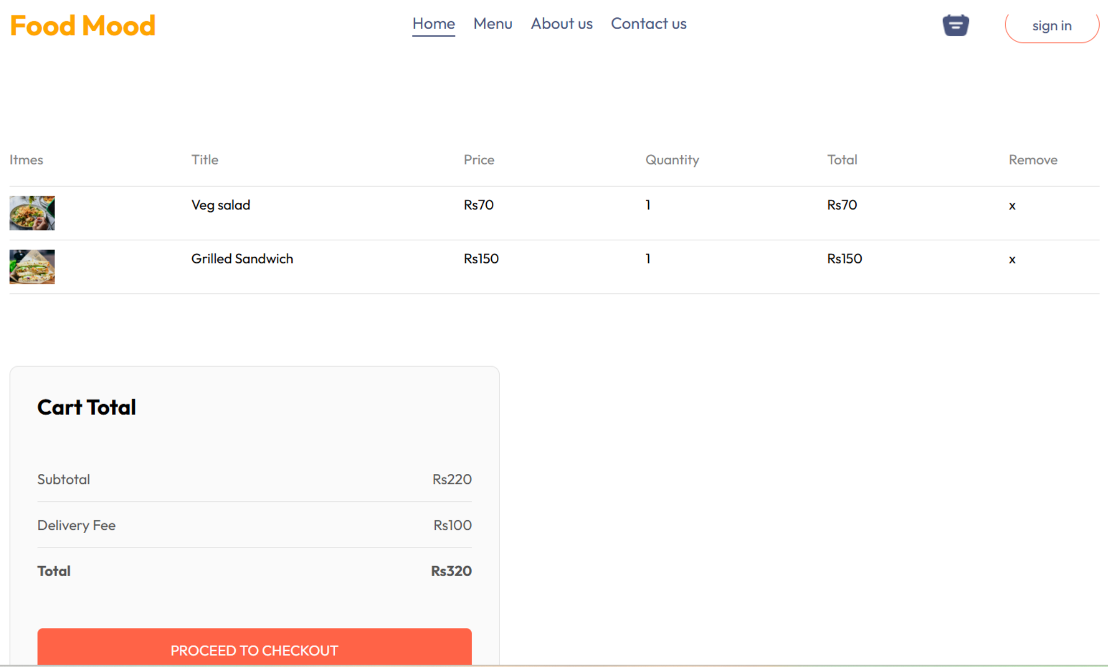
Cart management allows users to add, update, and remove food items before placing an order. The cart screen displays selected food items, quantity, price details, and total amount. It helps users manage their orders easily and provides a smooth ordering experience through an interactive and user-friendly interface.



4.1 Figure: Cart Screen

4.17: Module 5: Cart Details

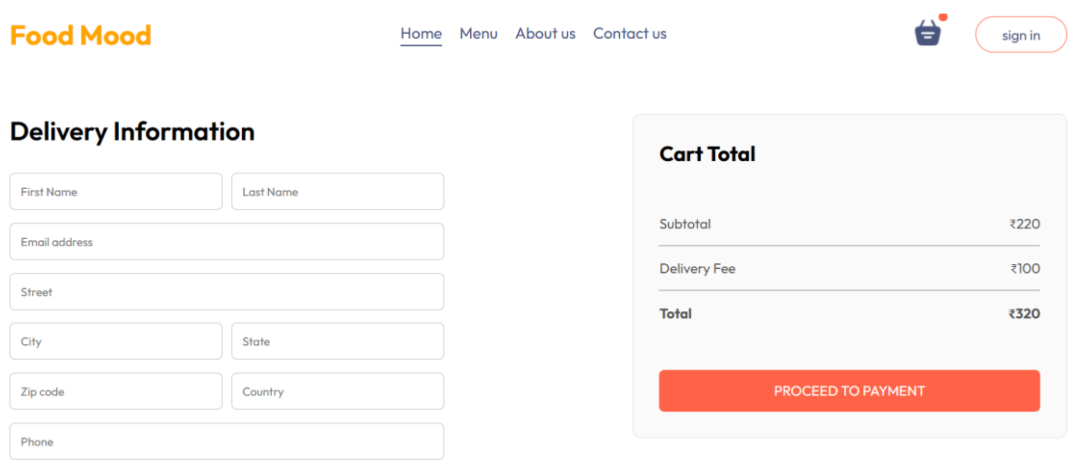
The cart details module displays all selected food items added by the user for purchase. The screen contains product information such as food name, image, quantity, price, subtotal, and total amount. Users can increase or decrease item quantity, remove products from the cart, and proceed to the checkout process for order placement.



5.1 Figure: Cart Details Screen

4.18: Module 6: Delivery Management

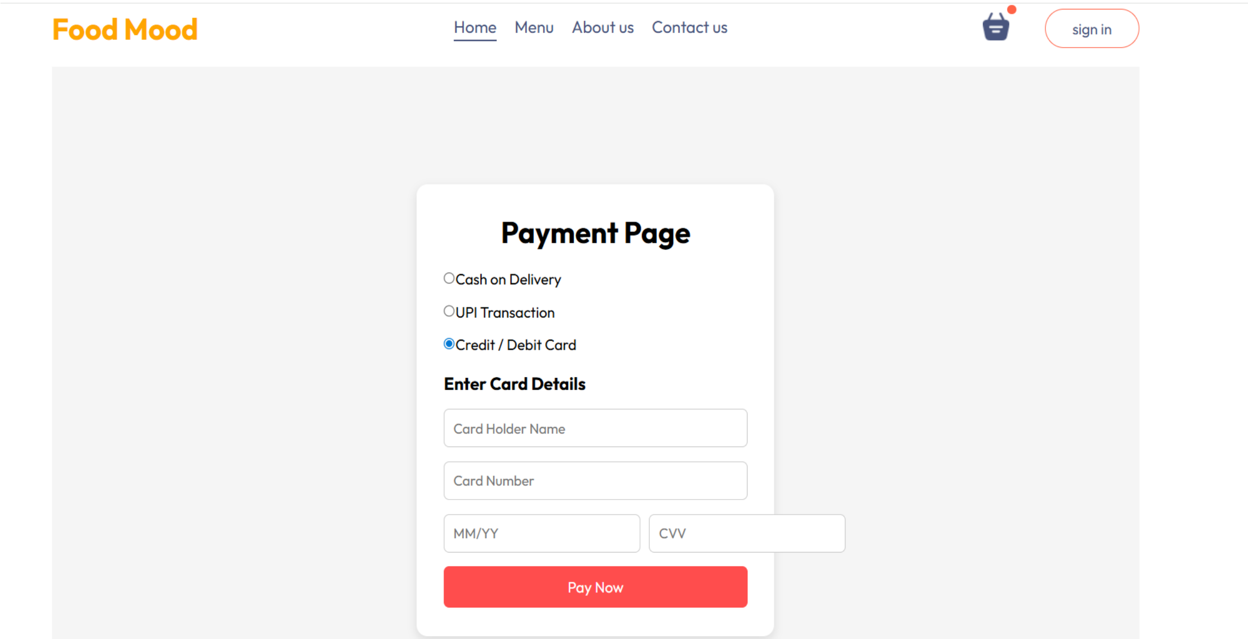
The delivery management module displays customer delivery address and cart total details for placed orders. It helps the admin verify delivery locations and monitor the total order amount during the order delivery process.



6.1 Figure: Delivery Managment Screen

4.19: Module 7: Payment System

The payment system module allows users to make secure online payments for food orders. The module processes payment details, verifies transactions, and confirms successful order payments. It helps provide a smooth, secure, and reliable payment experience for users during order placement



7.1 Figure: Payment System Screen

4.20: My Contribution to the Project

My main contribution to the Food Ordering System project was in MERN Stack web development using React.js, Node.js, Express.js, and MongoDB. I worked on frontend UI development, backend API integration, responsive webpage design, cart functionality, authentication system, and overall user interaction flow. I contributed to developing user pages, food item sections, cart management, order modules, payment integration, and admin dashboard functionalities.

The key contributions are listed below:

- Designed and implemented responsive React.js user interface screens for the Food Ordering System.
- Created reusable components for Navbar, Footer, Food Cards, Cart Sections, Buttons, and Order Pages.
- Worked on user authentication features including registration, login, and protected routes using JWT authentication.
- Developed cart management functionality including add-to-cart, quantity update, and remove item features.
- Created food item display pages and integrated dynamic product data using APIs.
- Assisted in frontend-backend integration using REST APIs and Axios for data communication.
- Verified and managed application data using MongoDB Compass and MongoDB collections.
- Improved UI responsiveness, layout consistency, form alignment, and overall user experience of the

4.21: Frontend Development Approach

The frontend of the Food Ordering System was developed using a modular and component-based approach in React.js. Instead of creating the entire application as a single page, the interface was divided into smaller reusable components such as Navbar, Footer, Food Item Cards, Cart Sections, Login Forms, Order Pages, and Admin Dashboard modules. This approach improved code organization, readability, reusability, and simplified future updates andmaintenance.

The application followed a consistent and responsive layout design for better user experience. The user interface included navigation menus, food display sections, cart management pages, payment pages, and order tracking screens.

The admin panel used dashboard-based navigation for managing food items, customer orders, and delivery status efficiently.

React Router DOM was used for page navigation, while React Hooks and state management were used for handling dynamic data and user interactions. Responsive design principles were followed using CSS3 and Bootstrap to ensure smooth accessibility across desktop, tablet, and mobile devices.

4.22: API Integration Approach

The frontend of the Food Ordering System was developed to communicate efficiently with backend APIs created using Node.js and Express.js. API integration was important because the frontend alone cannot permanently store, retrieve, or manage application data. In this project, API calls were required for user authentication, food item management, cart operations, order placement, payment processing, and order tracking.

During API integration, request and response data were checked carefully to ensure proper communication between the frontend and backend. Axios was used in React.js for sending HTTP requests and handling API responses dynamically. Postman and browser developer tools were used to test APIs and verify whether correct data was being sent and received successfully.

This process helped in identifying and resolving issues related to incorrect API routes, missing request fields, authentication errors, database connectivity, and response handling. API integration improved the overall functionality, responsiveness, and reliability of the Food Ordering System.

4.23: Authentication and Role-Based Flow

The Food Ordering System uses a secure authentication and user flow mechanism. After successful login, users can access application features such as browsing food items, managing carts, placing orders, and tracking order details. The admin can access the dashboard for managing food items, customer orders, and delivery status. This separation is important because users and admin should have different permissions and access levels within the application.

JWT authentication was used to maintain secure access to the system. After successful login or registration, a token is generated and used for protected API requests. This helps prevent unauthorized access to restricted modules and improves the overall security structure of the application.

4.24: User Interface and User Experience Considerations

The user interface of the Food Ordering System was designed to be simple, attractive, and user-friendly. Important features such as Login, Add to Cart, Place Order, Payment, Order Tracking, and Logout were placed in visible and accessible positions for better user interaction. Food items were displayed using cards with images, prices, and descriptions to provide a clear and interactive browsing experience

Proper spacing, alignment, navigation flow, button positioning, color consistency, and responsive layouts were considered while designing the application screens. The cart and order sections were structured clearly so users could easily manage food items and complete orders without confusion.

A clean and responsive user interface is important in online food ordering applications because users expect fast navigation and smooth interaction while browsing menus and placing orders. The use of React.js components and responsive design techniques improved the overall user experience across desktop, tablet, and mobile devices.

4.25: Testing and Debugging Strategy

Testing was performed module by module in the Food Ordering System. Initially, the frontend screens were checked for proper layout, responsiveness, and alignment. After that, user functionalities such as login, food browsing, cart operations, order placement, and payment processing were tested carefully. Later, frontend-backend communication and database operations were verified through API testing and MongoDB validation.

Table 4.6: Testing Activities Performed

Testing Area	Testing Activity
Login and Registration	Checked user registration, login validation, password handling, and error message display.
Home Page	Verified food item display, category filtering, navigation, and responsive layout behavior.
Cart Management	Tested add-to-cart, remove item, quantity update, subtotal, and total amount calculations.
Order Placement	Verified order confirmation, address handling, and order detail generation.
Payment System	Checked payment processing flow and successful payment confirmation handling.
Admin Dashboard	Verified food item management, order monitoring, and delivery status update functionality.
API Integration	Tested frontend-backend data flow, API responses, and request handling using Postman and browser developer tools.
Database Verification	Checked MongoDB collections for users, food items, carts, orders, and payment records.
Responsiveness	Verified application layout and UI behavior on desktop, laptop, tablet, and mobile screen sizes.

4.26: Documentation Work

Along with project development, documentation was also an important part of the internship. 2Screenshots of application modules such as login pages, food item sections, cart pages, payment screens, and admin dashboard were collected during the development process. Weekly activities, technologies learned, project modules, and implementation details were documented in a structured manner.

4.27: Food Managment Module Integration Flow

The Food Ordering System contains multiple connected modules that work together through the application flow. The authentication module provides secure login access for users and admin. After login, users can browse food items, manage carts, place orders, and make payments, while the admin can manage food items and customer orders through the dashboard.

The food item module is connected with the cart module, and the cart module is connected with order and payment functionalities. MongoDB stores all application data such as users, food items, carts, and orders.

This integration flow helped in understanding how different modules communicate with each other to provide smooth system functionality and better user experience.

Table 4.8: Food Managment Module Integration Flow

Module	Connection with Other Modules
Login and Authentication	Connects users and admin to secure dashboards and protected application pages.
Home and Food Items	Displays available food items and connects with cart management functionality.
Cart Management	Stores selected food items and connects with order placement and payment modules.
Order Management	Processes customer orders and connects with payment and delivery management.
Payment System	Handles payment processing and updates order status after successful transactions.
Admin Dashboard	Connects food item management, customer orders, and delivery status monitoring.
MongoDB Database	Maintains collections for users, food items, carts, orders, payments, and admin records.

4.28: Data Handling and Validation Approach

Data handling is an important part of the Food Ordering System because the application manages user accounts, food items, cart details, orders, and payment information. During implementation, input fields and forms were checked carefully to ensure that users could enter and manage data properly. The screens were designed with clear labels, placeholders, and organized layouts for better usability.

Validation and testing were also important while developing the application. Empty fields, incorrect inputs, invalid routes, and data flow issues were checked repeatedly during testing. MongoDB Compass was used to verify whether user, cart, order, and payment data were stored correctly in the database collections. This process improved understanding of frontend-backend communication and complete data flow in the MERN Stack application.

The main data handling points considered during the project were:

- Keeping user, food item, cart, order, and payment data in separate MongoDB collections.
- Checking input field labels, placeholders, and form validations for better user understanding.
- Verifying stored records using MongoDB Compass.
- Testing whether API response data appears correctly on the frontend screens.
- Maintaining secure access and proper separation between user and admin functionalities.

5: Findings and Recommendations

5.1: Findings from Internship

The internship helped me understand that real-world MERN Stack development requires proper planning, frontend-backend integration, database management, testing, and teamwork along with coding skills. While working on the Food Ordering System, I observed that a responsive and user-friendly interface improves user experience and makes application usage easier and faster.

The following findings were observed during the internship:

- Component-based development in React.js makes the application modular, reusable, and easier to maintain.
- Proper routing and navigation are important for smooth user interaction in multi-page web applications.
- API integration plays an important role in connecting frontend functionalities with backend services and database operations.
- MongoDB is suitable for storing flexible and dynamic application data such as users, food items, carts, and orders.
- JWT authentication improves application security by providing token-based access control for users and admin.
- Regular testing and debugging of modules help reduce integration and runtime issues during project development.

5.2: Suggestions for Improvement

Based on the internship experience the following constructive suggestions can be considered for further improvement of the internship program and learning process:

- More live project sessions and practical implementation activities can be conducted to improve hands-on industry experience.
- Regular code review and feedback sessions can help interns improve coding standards and project quality.
- Additional workshops on advanced MERN Stack concepts, deployment, and cloud technologies can enhance technical learning.
- A structured task management and progress tracking system can help interns manage assignments more effectively.
- More industry-level project exposure and team collaboration activities can help interns better understand real software development workflows.

5.3: Professional Learnings

The internship improved my understanding of professional discipline, task responsibility, and time management. I learned how to complete assigned work, ask doubts, test the implementation, and improve the output based on feedback. It also helped me gain confidence in frontend development and project explanation.

5.4: Project Results

The Project was successfully developed with major modules required for online food ordering and management. The frontend screens were designed with a clean, responsive, and user-friendly interface. Features such as user authentication, food item display, cart management, order placement, payment system, delivery management, and admin dashboard were implemented successfully.

The project achieved proper frontend-backend integration using REST APIs and MongoDB database connectivity. The application was tested for functionality, responsiveness, and smooth user interaction, which helped improve the overall performance and usability of the system.

5.5: Functional Outcomes

The following functional outcomes were achieved:

- Users can register and log in securely using authentication functionality.
- Users can browse food items with images, prices, and descriptions.
- Users can add food items to the cart, update quantities, and remove items from the cart.
- Users can place food orders and proceed with online payment functionality.
- Admin can manage food items, monitor customer orders, and update delivery status through the dashboard.
- User, food item, cart, order, and payment data are stored and managed using MongoDB database collections.

5.6: Learning Outcomes

The internship and project helped me gain practical knowledge in the following areas:

- React.js component-based frontend development.
- MERN Stack application architecture and workflow.
- User authentication and protected route handling using JWT.
- API integration and frontend-backend communication.
- MongoDB database collections and verification using MongoDB Compass.
- Git and GitHub workflow for version control and project management.
- Responsive UI design, debugging, testing, and project documentation.
- Cart management, order placement, and payment system implementation.

5.7: Challenges Faced

While developing the Food Ordering System, some challenges were faced:

- Maintaining responsive and consistent UI design across multiple screens.
- Understanding frontend-backend data flow during API integration.
- Handling authentication and protected route functionality using JWT.
- Managing cart operations and dynamic order data.
- Debugging routing, API response, and form validation issues.
- Understanding MongoDB document structure for users, orders, and payment records.
- Resolving frontend errors and backend connectivity issues during development.

5.8: Solutions Applied

The challenges were solved by dividing the project into smaller modules, testing each functionality individually, using reusable React.js components, checking API responses carefully, and verifying stored data using MongoDB Compass. Frontend errors, routing issues, and API integration problems were corrected through continuous testing and debugging.

5.9: Module-Wise Outcome Mapping

The project outcomes can be mapped with the modules developed during the internship. This helped in verifying whether the major requirements were covered in the final Food Ordering System implementation.

Table 5.1: Module-Wise Outcome Mapping

Module	Expected Outcome	Achieved Outcome
Login Module	Secure login access for users and admin.	Login and authentication flow were implemented successfully.
Food Item Module	Display food items dynamically.	Food item listing and category-based display screens were developed.
Cart Management	Manage selected food items before ordering.	Add-to-cart, quantity update, and remove item functionalities were implemented.
Order Module	Place and manage customer orders.	Order placement and order detail functionalities were developed.
Payment System	Support secure online payment.	Payment integration and payment confirmation flow were implemented.
Admin Dashboard	Manage food items and customer orders.	Admin dashboard for food and order management was created.

Module	Expected Outcome	Achieved Outcome
Delivery Management	Display delivery details and order total.	Delivery address and cart total management screens were developed.
Database Verification	Store and verify application data.	MongoDB Compass was used to verify users, food items, carts, and orders collections.

5.10: Benefits of the Developed System

The developed Food Ordering System provides several benefits from both user and technical perspectives. It simplifies online food ordering, improves customer convenience, reduces manual order handling, and provides organized data management for food items and customer orders. The system also supports secure authentication, cart management, online payment, and delivery tracking features.

From a technical perspective, the project helped demonstrate the use of React.js for frontend development, Node.js and Express.js for backend APIs, MongoDB for database management, and JWT for authentication. The modular structure of the project also makes it suitable for future enhancements and feature additions

5.11: Self Evaluation

During the internship, I improved my ability to understand project requirements and convert them into responsive MERN Stack application modules. I also improved my confidence in developing frontend interfaces, integrating APIs, managing MongoDB collections, and explaining project workflow and functionality.

The internship helped me identify areas where I need to continue improving, such as advanced backend development, deployment processes, performance optimization, and advanced testing techniques. The overall experience improved both my technical knowledge and practical software development skills.

5.12: Overall Professional Growth

The internship helped me grow professionally by providing exposure to real-world software development practices and project workflows. I learned that a developer should focus not only on coding but also on responsiveness, maintainability, user experience, testing, debugging, and proper documentation. I also understood the importance of teamwork, communication, time management, and accepting feedback positively for continuous improvement.

Working on the Food Ordering System improved my confidence in technical communication and project explanation. I became more comfortable explaining frontend-backend integration, API flow, MongoDB database collections, authentication processes, and application modules. This professional growth will be beneficial for future software development roles, technical discussions, and interviews.

The important improvements observed during the internship were:

- Better understanding of React.js component structure and reusable frontend development.
- Improved confidence in understanding project requirements and application workflow.
- Better understanding of API integration and frontend-backend communication.
- Improved debugging and testing skills for UI screens and application functionalities.
- Better understanding of MongoDB database verification using MongoDB Compass.
- Improved ability to prepare technical documentation and explain project implementation.

6.Conclusion

6.1: Conclusion

The Industrial Internship Program at ScaleFull Technologies LLP, Pune provided valuable practical experience in MERN Stack web development. During the internship, I worked on the development of a Food Ordering System using React.js, Node.js, Express.js, and MongoDB.

The project helped me understand frontend development, backend API integration, authentication flow, database management, testing, debugging, and project documentation. I learned how real-world web applications are planned, developed, tested, and managed using a modular approach. The internship improved my technical skills, confidence, communication abilities, and professional understanding of software development practices.

The Food Ordering System successfully includes important modules such as user authentication, food item management, cart management, order placement, payment system, delivery management, admin dashboard, and MongoDB-based data storage. The project can be further enhanced and deployed as a complete production-ready web application.

6.2: Future Scope

- The Food Ordering System can be enhanced in the following ways:
- Adding real-time order tracking and notification features.
- Adding customer reviews and food rating functionality.
- Implementing advanced analytics dashboard for admin reports and sales monitoring.
- Supporting multiple payment gateways and digital wallet integration.
- Deploying the application on cloud platforms such as AWS, Render, or Vercel.
- Improving mobile responsiveness and accessibility features.
- Adding email and SMS notifications for order confirmation and delivery updates.
- Implementing advanced security and automated testing features.

6.3: Project Limitations

The current version of the Food Ordering System mainly focuses on core functionalities such as user authentication, food item management, cart operations, order placement, payment handling, and delivery management. Some advanced features such as real-time notifications, customer reviews, detailed analytics, cloud deployment, and automated testing can be added in future versions.

The system can also be improved further by implementing advanced security features, better scalability support, and additional payment and delivery tracking functionalities.

6.4: Final Summary

The internship and project work helped me improve my practical knowledge in Full Stack Web Development using MERN Stack technologies. During the internship at ScaleFull Technologies LLP, I worked on frontend and backend development tasks, API integration, authentication systems, database connectivity, responsive UI design, testing, debugging, and project documentation.

The internship provided hands-on experience in developing real-world web applications using React.js, Node.js, Express.js, and MongoDB. I also learned about reusable components, routing, state management, GitHub version control, and project structure. Overall, the Industrial Internship Program enhanced my technical skills, problem-solving ability, and confidence, and prepared me for professional software development roles in the IT industry.

6.5: Final Internship Reflection

This internship was highly beneficial because it connected my academic learning with practical software development experience. Before the internship, most of my knowledge was limited to classroom concepts, academic assignments, and small practice projects. During the internship, I understood how industry-level web applications are planned, designed, developed, tested, and maintained using modern technologies.

Working on MERN Stack projects helped me gain practical exposure to frontend development using React.js and backend development using Node.js and Express.js. I learned how APIs are integrated, how MongoDB databases are connected and managed, and how authentication systems and dynamic user interfaces are implemented. The internship also improved my understanding of debugging, testing, responsive web design, and code organization.

Another important learning experience was project documentation. Preparing the internship report helped me organize and present the complete project work in a structured format including objectives, technologies used, implementation details, weekly progress, challenges faced, testing, results, and future enhancements. This improved my technical writing and presentation skills.

Overall, the internship improved my confidence as a MERN Stack Developer and gave me valuable industry exposure. It motivated me to continue learning advanced web technologies, backend integration, deployment processes, and professional software development practices that will support my future career growth in the IT field.

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